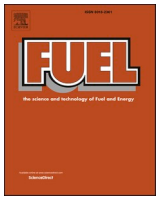
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Hydrolysis and methane yield enhancement of anaerobic co-digestion of kitchen waste and sewage sludge by air/carbon dioxide nanobubble water

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ABSTRACT

Anaerobic co-digestion of kitchen waste and sewage sludge has attracted considerable attention, but the accumulation of volatile fatty acid can affect biogas yield and hinder its application. In this study, the effects of air/carbon dioxide nanobubble water (NBW) on hydrolysis and methane yield were investigated. Under mesophilic conditions, the addition of different NBW (Air, CO_2, and Air + CO_2) increased the total ammonia nitrogen in the hydrolysis stage by 6.6 %, 11.5 %, and 7.0 %, respectively. Meanwhile, the volatile fatty acid ratio total alkalinity of NBW_{Air+CO_2} was lower than that of the other NBW digesters, improving system stability. In addition, NBW increased ultimate methane yield by 4.7 %, 25.3 %, and 21.0 %, respectively. The addition of NBW also affected the microbial diversity and relative abundance of microorganisms in the digesters. The alpha diversity index of NBW_{Air+CO_2} is superior to other NBW digesters. Bacteria enriched with *Firmicutes*, *Bacteroidetes*, *Proteobacteria,* and *Chloroflexi* improved the hydrolysis, and methanogens were dominated by hydrotrophic types. In summary, the introduction of NBW_{Air+CO_2} provides new insights to improve hydrolysis, system stability, and methane yield efficiency.

1. Introduction

Kitchen waste (KW) in China is expected to exceed 103.0 million tons by 2025 [1]. The large amount of KW has caused a series of environmental problems such as water pollution, garbage accumulation, odor pollution, etc., which need to be properly treated [2]. Anaerobic digestion (AD) is a cost-effective strategy to convert organic wastes such as KW into biogas for energy recovery [3]. However, when KW is the only substrate, the long-running AD process is susceptible to volatile fatty acid (VFA) inhibition, leading to poor performance or even failure of the AD [4]. This is mainly caused by the high carbon ratio nitrogen (C/N) of KW, and in this regard, anaerobic co-digestion has been proposed to increase methane yield by adjusting C/N to an optimal level and supplementing deficient nutrients or trace elements [3].

Sewage sludge (SS) is a possible co-digestion substrate with KW since it has a lower C/N and slower hydrolysis rate. The efficiency of anaerobic co-digestion was higher than that of pure SS or pure KW at appropriate mixing ratios. At a 1:4 volatile solid (VS) ratio of KW and SS, methane yield was increased by 85.3 % compared with the digestion of KW alone [5]. In addition, higher methane recovery was observed at SS to KW mixing ratio of 1:1 (based on the dry VS), which resulted in a 4.6 times higher methane yield, and a 3.9 times higher hydrolysis rate compared with SS mono-digestion [6]. Overall, the successful co-digestion of KW and SS confirms the superiority and the existence of synergistic effects that can accelerate the biodegradation process and recover methane more efficiently. Although co-digestion of KW and SS has attracted widespread focus, its further application still faces various unavoidable challenges, with a long lag phase, unstable performance, accumulation of VFA, and poor methane yield, which still need to be further optimized [1].

To address the above mentioned issues, many methods are available to increase methane yield and improve AD performance, such as pretreatment [7], biofortification [8], or digester optimization [9], which can be effective in mitigating inhibition. Even though the performance of AD is somewhat improved by these methods, the significant amount of chemicals added to AD increases the burden of ultimate waste treatment, which may be too expensive on an industrial scale when combined with the development of secondary pollutants [10]. Consequently, it is

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worthwhile to investigate an efficient and cost-effective solution strategy to improve the stability and performance of co-digestion of these two substrates. Water is crucial in the AD process, not only as a major component of the digestive fluid but also to facilitate the transfer of hydrolysis products to sites where they can be utilized by bacteria [11]. However, relatively few studies have been conducted on the effects of different types and characteristics of water on AD performance.

Nanobubble water (NBW), as a green and pollution-free additive, has been used to enhance the AD performance of organic waste [12]. NBW has a large number of nano-sized bubbles [13]. Due to its large size of specific surface area, and small density, NBW has a superior sorption capacity and prolonged storage time that ordinary water does not have, which is enough to improve the reaction speed between gas and liquid and increase the activity of the organisms in the water [14]. The addition of NBW_{CO_2} to the AD of corn stover resulted in the highest cumulative methane yield of 132.2 mL g^{-1} VS_{added}, which was 17.0 % higher than that of deionized water (DW) [15]. Meanwhile, the addition of NBW_{CO_2} accelerated the production and consumption of VFA. The addition of NBW_{Air} increased H_2 and methane yield by 38.0 % and 24.0 % in the two-phase AD of KW [16]. These results demonstrate that different NBW gas species were utilized to promote the degradation of various substrates to enhance biogas yield.

Each NBW gas has unique properties that can produce different effects in the AD process. NBW_{Air} addition may not only increase hydrolase activity at the end of the hydrolysis/acidification phase by inducing a microaerobic environment but also coenzyme activity at the end of the methanogenesis phase of the two-phase AD [17]. The addition of NBW_{CO_2} to the digesters increased the production of hydrogenotrophic methanogens, which in turn increased biogas yield [12]. In addition, CO_2 can increase microbial diversity by generating bicarbonate, altering the community structure of methanogens [18]. NBW_{Air} hybrid NBW_{CO_2} may provide both O_2 and CO_2, which promotes microbial metabolic activities and accelerates the degradation of organic matter. However, there are few studies on the effects of the hybrid application of NBW with these two gases on the AD process, and further research on this issue is needed to fully realize the potential for enhanced substrate degradation and overall biogas yield.

Therefore, this study investigated the effect of NBW_{Air}, NBW_{CO_2}, and their hybrid (NBW_{Air+CO_2}) on the co-digestion of KW and SS; the performance with or without NBW addition was evaluated by measuring the general parameters and observed the differences in methane yield and microbial community.

2. Materials and methods

2.1. Substrate and inoculum

KW was collected from a university canteen in Lanzhou City, China. The collected KW was pretreated by sorting out plastic, bones, tissues, chopsticks, etc. The KW was then crushed and mixed, and the resulting paste was passed through a 0.25 mm sieve for spare use. SS was taken from a wastewater treatment plant in Lanzhou City, China, and the collection object was originally dewatered sludge, using the "oxidation ditch + aerated biofilter process", with a scale of 5000.0 m^3 d^{-1}. The SS was recovered and domesticated in mesophilic (37 ± 2 °C) conditions and then used as inoculum. The characteristics of KW, SS, and inoculum are shown in Table 1.

2.2. Preparation of nanobubble water

The NBW in this study was prepared using a nanobubble generator (JXWNP-AL01, Jinxiang, China). Air and CO_2 were introduced into DW via the generator, kept at 20 ± 2 °C for 20 min, and then left to stand for 10 min to obtain NBW for use throughout the AD process. The pressure of the generator was controlled between 0.4–0.5 MPa during NBW preparation. Once prepared, the NBW is used in the ensuing AD process.

Table 1
Characteristics of kitchen waste, sewage sludge, and inoculum.

	Kitchen waste	Sewage sludge	Inoculum
TS (%)	24.18 ± 0.84	17.48 ± 0.22	6.12 ± 0.66
VS (%)	22.93 ± 0.70	8.23 ± 0.29	2.55 ± 0.08
VS/TS (%)	93.22 ± 0.19	45.84 ± 0.50	46.75 ± 0.76
C (%)	50.41 ± 0.36	26.06 ± 0.02	23.73 ± 0.15
N (%)	2.91 ± 0.05	2.59 ± 0.01	2.83 ± 0.04
H (%)	6.98 ± 0.08	3.62 ± 0.04	4.07 ± 0.08
S (%)	0.17 ± 0.01	0.64 ± 0.04	0.73 ± 0.01
C/N	17.34 ± 0.41	10.05 ± 0.03	8.40 ± 0.16
pH	5.71 ± 0.02	7.67 ± 0.03	7.95 ± 0.02
SCOD (mg/L)	56621.99 ± 136.25	8368.20 ± 68.12	5598.37 ± 34.06

Note: Determination of C, N, H, and S content are all based on the dry matter of the kitchen waste and sewage sludge.

2.3. Anaerobic digestion experimental setup and Operation

The experimental setup consisted of a plexiglass digestion flask, a gas collection bag, a gas tube, a sampling tube, a filter head, and a water bath. The volume of the digester was 1 L. The digester setup used is the same as Liu et al [19]. To ensure smooth sampling, a filter head was connected to the inner part of the sampling metal tube to prevent sampling clogging, and the outer part of the tube was used for sampling and testing various parameters.

Based on the preliminary experiment, the optimum substrate mixing ratio was selected by mixing KW and SS in a 1:2 (VS) ratio, and 406 g was taken as substrate. Add inoculum to each digester with a ratio of inoculum to working volume of 1:5. With 261.0 g of DW as control, while 261.0 g of different types of NBW were added to a 1L digester as experimental groups with a total volume of 800 mL. The digesters were placed in a constant water bath at a temperature of 37 ± 2 °C for batch AD experiments. The four groups in this study were: the addition of DW, the addition of NBW_{Air}, the addition of NBW_{CO_2}, and the addition of NBW_{Air+CO_2}. All experimental groups had four parallels. The bottles were gently shaken manually every day to ensure that the reaction was complete. On days 0, 3, 7, 14, 21, 29, and 40, a peristaltic pump was used to extract about 30.0 g of digestate for analysis.

2.4. Methods of analysis

The digestate is centrifuged at 2000 *g* for 10 min and the liquid supernatant is strained through a 0.45 μm polyethersulfone ultrafine fiber filter. The parameters are determined using the strained liquid. Total solid (TS), VS, soluble chemical oxygen demand (SCOD), total ammonia nitrogen (TAN), and total alkalinity (TA) were determined using standard methods [20]. The pH was determined using a handheld high precision pH meter. The meter was calibrated with a standard solution before use and readings were recorded after stabilization. Free ammonia nitrogen (FAN) in liquid samples is measured by a standard formula [21]. Concentrations of VFA (acetic, propionic, isobutyric, n-butyric, isovaleric, and n-valeric acids) were determined by using a gas chromatograph (SP-7890 Plus, Rui Hong, China) equipped with an organic acid analytical column (30.0 m long, 320.0 μm internal diameter, 0.5 μm film) and a hydrogen flame ionization detector. The amount of biogas was determined by a gas flow meter (LML-2, Alpha, China). Methane concentration in the biogas was assessed using a biogas analyzer (GA5000, Geotach, UK).

Deoxyribonucleic acid was extracted from digestate, and the microorganisms contained therein were tested for abundance and type using high-throughput sequencing to determine differences in the microbial community in the digesters after the addition of different NBW. Primers 341F (5′-CCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGGTWTCTAAT-3′) were utilized to amplify the bacterial 16S rRNA gene in the V3-V4 portion of the bacterial 16S rRNA gene; using primers Arch519F (5′-CAGCCGCCGCGGTAA-3′) and Arch915R (5′-

GTGCTCCCCCGCCAATTCCT-3′) to amplify the V4 portion of the archaeal V4-V5 portion of the 16S rRNA gene. Each sample was subjected to DNA extraction, followed by sequencing, and finally by processing the raw data and species information to generate complete operational taxonomic units (OTUs). The original data were uploaded to NCBI with accession numbers PRJNA1083726 and PRJNA1083733.

2.5. Statistical analysis

Data were analyzed using SPSS for Windows system version 27.0. Data between different experimental groups were analyzed using ANOVA (analysis of variance) and differences between two experimental groups were considered to exist when $p < 0.05$. In addition, principal component analysis (PCA) and Duncan's test were performed. Methane yield is fitted using a modification of the Gompertz model. Spearman correlation analysis was performed between microbial relative abundance (RA) and environmental factors in the experiment.

3. Results and Discussion

3.1. Digestion performance

The pH is a key factor in the AD process and directly affects the stability of the AD system [15]. During the start of the AD process, a rapid decrease in pH occurs in all digesters owing to the amount of acid during the acid production phase. This second decrease can be attributed to the organic matter released by the microbial cells in an alkaline environment [22]. Throughout the reaction phase, the pH is maintained between 6.5–7.8, within the normal range of the AD process (6.5–8.2) [23].

Typically, SCOD increases with hydrolysis and acidification and is partially removed as methanogens consume organic acids, converting them into methane and carbon dioxide, therefore the difference in SCOD is closely related to biogas yield [14]. The SCOD in different experimental groups showed a similar trend of increasing first and then decreasing (Fig. 1**b**). On day 3, the SCOD reached the maximum, which may have been caused by the release of a large amount of VFA during the hydrolysis stage. At this time, the SCOD in the NBW groups was higher than that in the DW group, indicating that the addition of NBW could promote the hydrolysis of co-digestion of KW and SS, and the highest SCOD increased by 5.7 %, 10.4 % ($p < 0.05$) and 8.5 % in the digesters of NBW_{Air}, NBW_{Air}, and NBW_{Air+CO_2}. After day 3, the SCOD began to decrease. It is noteworthy that the NBW_{Air+CO_2} groups had the highest rate of decrease in SCOD during this period, 79.9 %, which was superior to DW (78.7 %), NBW_{Air} (78.1 %), and NBW_{CO_2} (78.8 %), and this phenomenon suggests that the hybrid NBW facilitates the decomposition of organic matter. After day 29, the second increase in SCOD implies that the microorganisms ruptured in the highly alkaline environment, releasing organic matter [22].

TAN is a result of the hydrolysis of proteins and is used to reaction

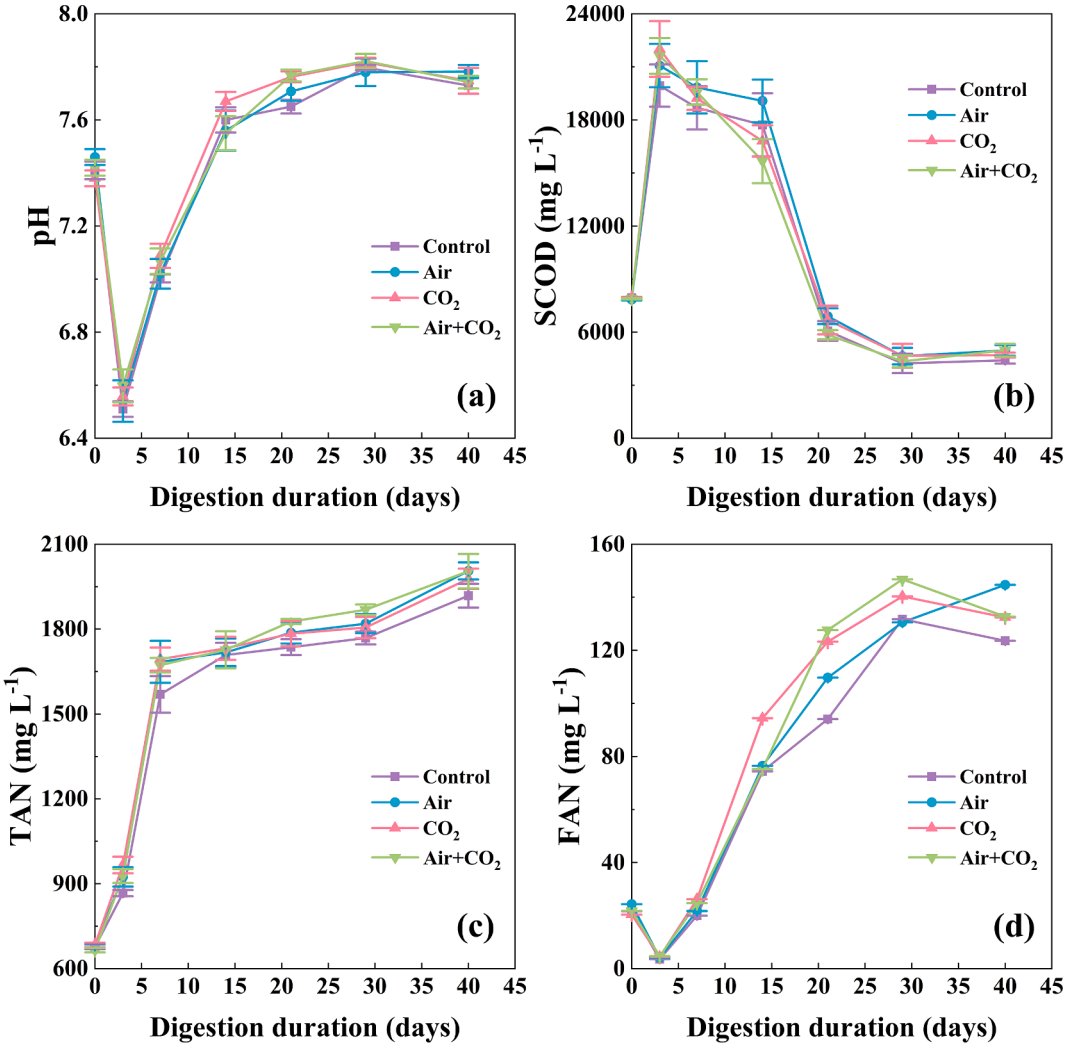


Fig. 1. Changes of pH (a), SCOD (b), TAN (c), and FAN (d).

the hydrolysis performance of AD [19]. Low levels of TAN can provide cushioning and nutrition for microbial, however high levels can inhibit the growth of methanogens [24]. In the early stage of the AD process, the TAN in each digester gradually increased and then stabilized (Fig. 1**c**). This may be due to the decomposition of proteins in the early stage of AD, which increases the amount of inorganic nitrogen such as ammonia nitrogen. TAN in all digesters reached 1918.0–2005.8 mg L^{-1} at the end of AD, suggesting that ammonia inhibition may be occurring in DW and NBW when the rate of methane yield slows [25]. The TAN in the experimental groups was higher than those in previous studies, which may be caused by different substrates and NBW addition to promote hydrolase activity [26]. On day 3, NBW_{Air}, NBW_{CO_2}, and NBW_{Air+CO_2} significantly increased TAN by 6.6 % ($p < 0.01$), 11.5 % ($p < 0.001$), and 7.0 % ($p < 0.001$), respectively, as compared with the DW, with significant differences between groups. This result suggests that TAN is more affected by different NBW during the hydrolysis stage. In addition, NBW_{Air} and NBW_{Air+CO_2} showed an increase of 4.6 % ($p < 0.05$) and 4.5 % ($p < 0.05$) on day 40, whereas NBW_{CO_2} (3.1 %) showed no significant difference. This suggests that NBW_{Air} and NBW_{Air+CO_2} played an equal role in the methanogenic stage, while NBW_{CO_2} did not play a significant role. This may be due to the fact that NBW_{Air} generates a large amount of reactive oxygen species, which improves the digestibility of the substrate and thus promotes the oxidation of organic matter [12].

FAN can penetrate microbial cell membranes and cause proton imbalances or toxic cytoplasmic enzymes [14]. Therefore, the toxicity of FAN is greater than that of ionic ammonia. As AD proceeded, FAN gradually increased (Fig. 1**d**). After day 29, the FAN in DW, NBW_{CO_2}, and NBW_{Air+CO_2} decreased, which may be due to a decrease in pH. Since the pH of NBW_{Air+CO_2} is lower, although the TAN is higher than in other NBW digesters, the FAN is also lower. When FAN exceeds 250.0 mg L^{-1}, methanogens are affected [22]. In this experiment, the FAN in all digesters consistently remained below the threshold.

VFA are the products of acid generation and methanogenic substrates in the AD process. It is an essential index of the stability and function of the AD system [27]. The trend of VFA was similar to that of SCOD, both showing a trend of increasing and then decreasing (Fig. 2). The high concentration of VFA was almost completely consumed from day 7 to day 40 and the study showed a similar trend of VFA degradation [24]. Digesters VFA added with NBW accumulated more in the early stage, providing more feedstock for methanogens. On day 3, the VFA peaked and was mainly dominated by propionic acid, its higher concentrations (1000.0–2000.0 mg L^{-1}) are susceptible to methanogens [28]. At this time, NBW_{Air} had the highest concentration of VFA, which was inconsistent with the trend of SCOD, probably because some of the VFA in the NBW_{CO_2} and NBW_{Air+CO_2} digesters had already been utilized by methanogens and converted to methane [15]. As fermentation progressed, NBW_{Air+CO_2} had lower levels of acid accumulation in the VFA. These results suggest that the addition of hybrid NBW facilitates the consumptive use of VFA for methane conversion, which restores the stabilization of the AD system.

TA is related to the buffering capacity of the digesters [29]. As fermentation proceeded, the TA in each digester gradually increased, indicating enhanced buffering capacity (Fig. 3**a**). After day 3, the TA in all the digesters with NBW addition was higher than in DW, suggesting that the addition of NBW promotes the production of TAN and thus increases the TA, which may play an important role in restoring the stabilization of the acidic AD system.

The VFA ratio TA (VFA/TA) is already used in the AD system as a key indicator for evaluating the stabilization of the AD process, with ratios greater than 0.4 indicating an unstable AD process [14]. In the early stage, the VFA/TA of all digesters ranged from 1.9 to 2.0, which was higher than the threshold value, and DW was the lowest (Fig. 3**b**). This indicates that at the beginning of the AD process, the digesters were in an unstable state. As AD proceeded to day 7, the VFA/TA of all digesters decreased, with the NBW_{Air} and NBW_{CO_2} groups higher than the DW by 13.3 % and 9.3 %, while the NBW_{Air+CO_2} groups which were significantly lower than that of DW by 24.3 % ($p < 0.001$), and this change suggests that NBW_{Air+CO_2} can effectively promote the early recovery of digesters stability, which is important for alleviating the acid inhibition phenomenon during the AD process [26].

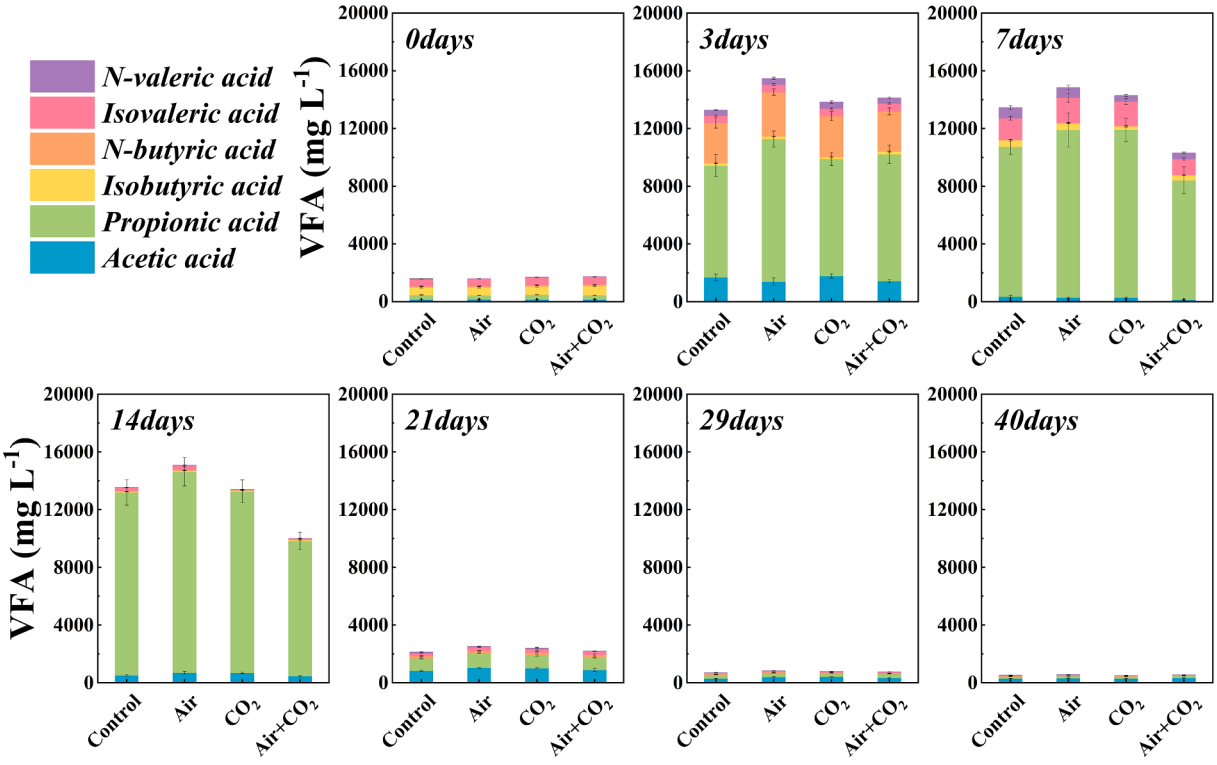


Fig. 2. Changes of VFA.

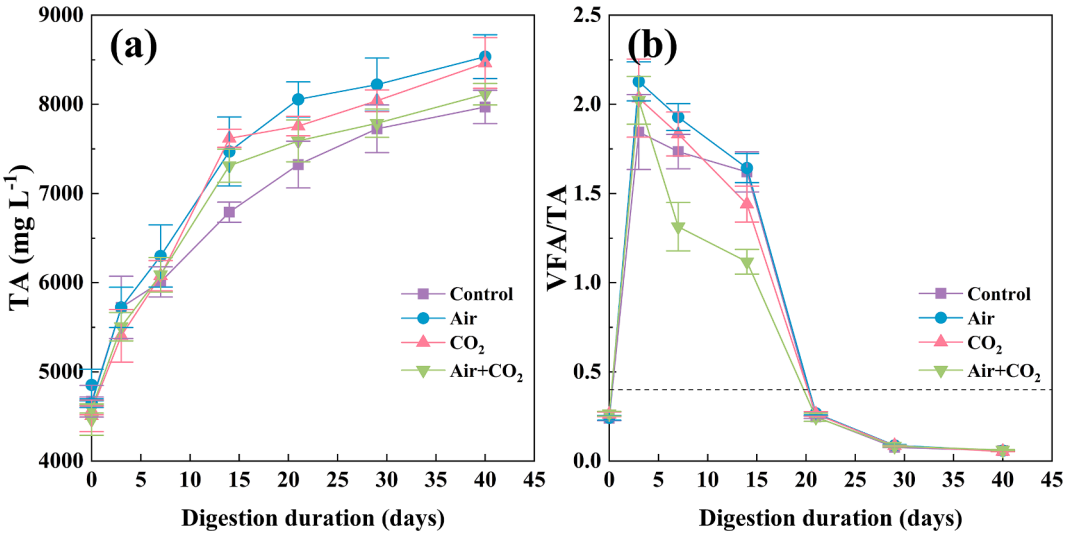


Fig. 3. Changes of TA (a) and VFA/TA (b).

3.2. Methane yield

The Gompertz model is suitable for co-digestion systems [30]. The model was used to fit the cumulative methane yield (CMY) and the results are shown for the lag phase (λ), maximum methane production rate (R_m), and ultimate methane yield (UMY) (Table 2). λ determines bacterial adaptation to the digester and biogas yield [31]. The NBW_{Air} λ was the shortest, however, only 246.78 ± 1.96 mL g^{-1} VS_{added} methane was recovered at the end. For NBW_{CO_2}, λ was about 3.8 d, which may be due to the adaptation of microorganisms to the supplement. NBW_{Air+CO_2} was adversely affected by 4.6 d, which was not different from the DW. A longer λ is acceptable if high biogas and methane yield is achieved [31].

Three peaks of R_m occurred in all groups (Fig. 4**a**). The first peak appeared on day 1–2 of co-digestion, and may be caused by the degradation of biodegradable organic matter [1]. The second peak occurred on day 4–7 and may be caused by the large amounts of acetic and butyric acids during methanogenesis. The last peak occurred on day 13–17 and may be caused by the transformation of propionic acid to acetic acid for use by methanogens [27]. During the early stage (day 0–2), the R_m increased from 0.0 to 7.5–9.3 mL g^{-1} VS_{added} d^{-1}. However, R_m was unstable from day 2 to day 8 due to the accumulation of VFA. After adapting to the environment, R_m increased steadily. It was also found that the digesters with DW and NBW_{Air} reached the highest R_m on day 16, whereas NBW_{CO_2} and NBW_{Air+CO_2} on day 14, suggesting that NBW_{CO_2} and NBW_{Air+CO_2} have a certain shortening effect on the startup of the digesters to speed up the adaptation of anaerobic microorganisms to the environment. NBW_{Air} is inconsistent with previous studies, which may be due to differences in substrates [16]. During AD, NBW_{Air+CO_2} had the largest R_m, which was significantly different from the DW, with an improvement of 39.1 % ($p < 0.01$).

The trend in CMY for digesters with NBW or DW added increased steadily with digestion duration (Fig. 4**b**). Overall, UMY was higher in NBW digesters than DW, and according to Duncan's analysis, the NBW groups were significantly different from DW. UMY increased by 4.7 % ($p < 0.05$), 25.3 % ($p < 0.001$), and 21.0 % ($p < 0.001$) in the digesters with the addition of NBW_{Air}, NBW_{CO_2} and NBW_{Air+CO_2}, respectively. NBW_{CO_2} and NBW_{Air+CO_2} were also significantly different from NBW_{Air}, being 19.7 % ($p < 0.001$) and 15.6 % ($p < 0.001$) higher than NBW_{Air} in both groups, whereas NBW_{CO_2} did not show a statistically significant difference from NBW_{Air+CO_2}. Where NBW_{Air} is due to the presence of a microaerobic environment and increased activity of hydrolyzing and acid-producing microorganisms in the digesters, which promotes substrate hydrolysis [17]. NBW_{CO_2} has the highest UMY probably since CO_2 is a readily soluble gas and more CO_2 in biogas can dissolve in water by forming NH_4HCO_3 under high pH and TAN conditions, which increases the methane production in the NBW_{CO_2} digesters [15]. In addition, CO_2 can be used as a substrate to promote the production of hydrogenotrophic methanogens using H_2 molecules as electron donors [12]. If the influence of removing CO_2 itself is to be considered, assuming complete conversion of CO_2, it can be calculated by the mass balance that the CO_2 added to the NBW_{CO_2} group theoretically converts 11.9 mL g^{-1} VS_{added} methane, yet NBW_{CO_2} was higher than DW 59.7 mL g^{-1} VS_{added} in this study, which suggests that the introduction of NBW_{CO_2} generates bicarbonate to increase microbial diversity and change the community structure of methanogens [15].

In the present study, several parallel experiments showed that the addition of NBW promoted UMY (Fig. 4) and digestion stability (Fig. 3**b**) during AD. NBW_{Air+CO_2} combines the advantages of the above mentioned two gases, which not only improves UMY performance but also provides a more cost-effective option for methane yield by using only half the cost of NBW_{CO_2}.

In the previous, methane yield increased by 20.8 % and 19.1 % for NBW_{Air} and NBW_{CO_2}, which was higher and lower than the present study [11]. This may be due to the different substrates. The addition of NBW_{Air} acted as a buffer, enhancing the buffering and resilience of the AD system and accelerating the production and consumption of reducing sugars and VFA [17]. Nanobubbles have high gas solubility, which facilitates the dissolution of more CO_2 and thus improves the buffering performance of the NBW_{CO_2} [15]. NBW increases water mobility in the surrounding aquatic environment, thereby enhancing microorganism activity. By increasing water mobility, the mass transfer of nutrients to

Table 2
Improved Gompertz model fitting results.

	Lag phase, λ (d)	Maximum methane production rate, R_m (mL g^{-1} VS_{added} d^{-1})	Ultimate methane yield, UMY (mL g^{-1} VS_{added})	R^2
DW	4.45 ± 0.56$^{b, c}$	15.46 ± 0.17^{a}	235.70 ± 5.37^{a}	0.974 ± 0.001
NBW_{Air}	3.69 ± 0.29^{a}	16.47 ± 0.41^{a}	246.78 ± 1.96^{b}	0.972 ± 0.002
NBW_{CO_2}	3.84 ± 0.26^{a}	21.36 ± 1.86^{b}	295.41 ± 4.80^{c}	0.976 ± 0.001
NBW_{Air+CO_2}	4.60 ± 0.01^{c}	21.51 ± 1.69^{b}	285.28 ± 8.11^{c}	0.975 ± 0.003

Note: a, b, and c are labeled according to the Duncan test, and differences between the two groups are considered significant at the $p < 0.05$ level when the symbols labeled between the two groups do not have repeated letters.

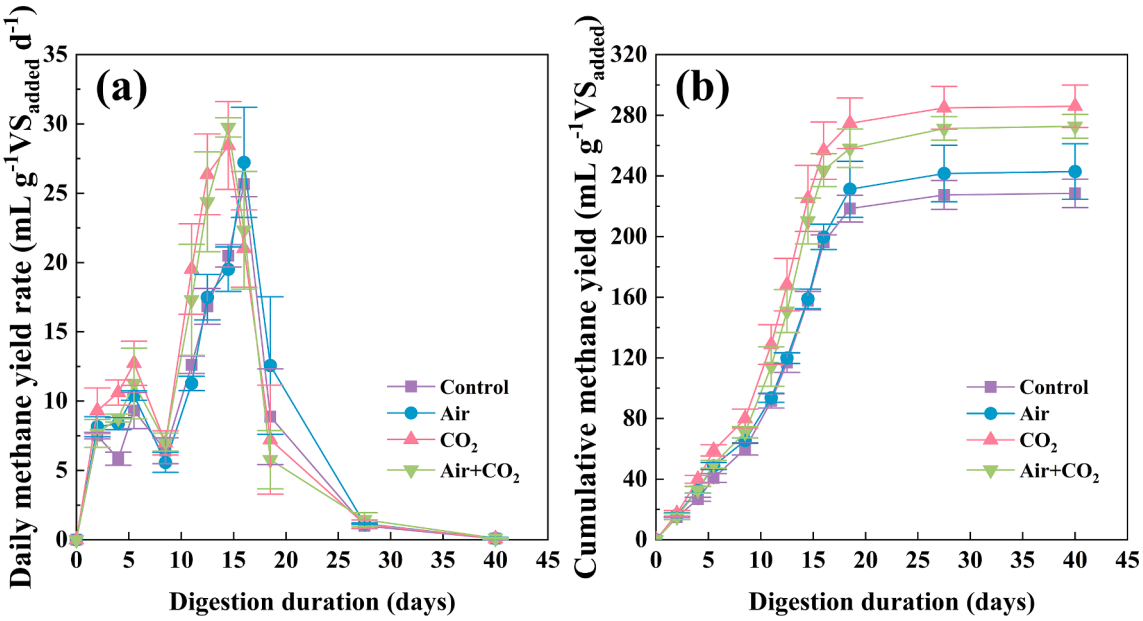


Fig. 4. Changes of cumulative methane yield (a), maximum methane production rate (b).

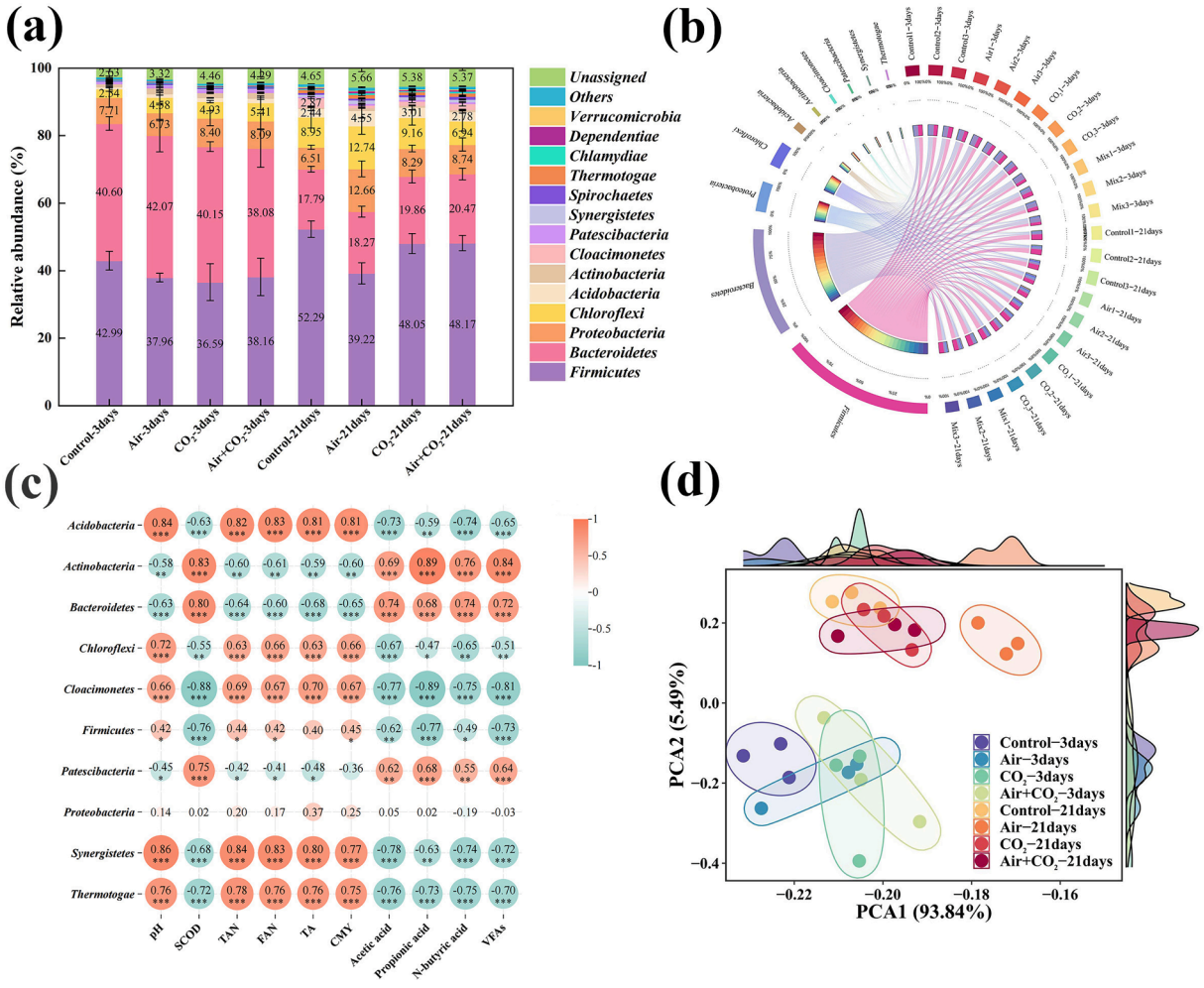


Fig. 5. Distribution of bacterial sequences at the phylum level, RA (a), Circos plot (b), Correlation analysis plots of environmental factors (c), and PCA plot (d).

microbial cells is increased, leading to increased enzyme activity, which in turn increases substrate decomposition rates and produces more biogas [12]. The NBW technique is not only capable of treating acidic substrates successfully but also eliminates the use of chemistry.

3.3. Characteristics of the microbial community

3.3.1. Diversity of microbial community

The OTUs, Shannon and Chao1 indexes of bacterial sequences were larger and Simpson indexes were smaller in all the digesters of this study, indicating that bacterial diversity and RA were much higher than archaea (**See supplementary material**). According to Duncan's test, the OTUs, Shannon, and Chao1 indexes of bacteria in digesters with NBW addition were significantly higher than DW during the methanogenic stage (day 21), which may be attributed to the higher delivery efficiency of NBW gas and its potential to penetrate the inner regions of bioflocs and biofilms due to its small size and relative stability, allowing them to be transported in the cells [32]. Among the three NBW, the OTUs, Shannon, and Chao1 indexes of the bacterial sequences in the digesters with NBW_{Air+CO_2} added were higher than those of the other NBW, suggesting that the different gaseous NBW also affected the species richness of bacteria.

3.3.2. The distributions of bacterial sequences

Bacteria were dominated at the phylum level by *Firmicutes*, *Bacteroidetes*, *Proteobacteria*, and *Chloroflexi* (Fig. 5**a** and Fig. 5**b**). During the methanogenic stage (day 21), the RA of *Firmicutes* increased and *Bacteroidetes* decreased in all digesters.

Firmicutes can decompose a wide range of complex organic matter, including proteins and carbohydrates, and are active in both mesophilic and thermophilic conditions [31]. The RA is the highest in the early and late stages of AD. It ranges from 36.6 %-43.0 % on day 3 and increases to 39.2 %-52.3 % on day 21. *Firmicutes* can form endospores, allowing them to endure unfavorable external conditions and survive in extremely harsh environments [33]. On day 3, the accumulation of VFA is severe in all digesters, which is not conducive to the multiplication and metabolism of microorganisms. However, *Firmicutes* adapt and become the dominant microorganisms in the community. After correlation analysis of bacteria with environmental factors, there is a negative correlation between *Firmicutes* and VFA ($p < 0.001$, R = -0.73), SCOD ($p < 0.001$, R = -0.76) (Fig. 5**c**). Compared with the DW, the RA of *Firmicutes* in the groups with added NBW is lower, indicating that the addition of NBW can alleviate the acidic inhibition in the digesters. The RA of *Firmicutes* grew faster in NBW_{CO_2} and NBW_{Air+CO_2} compared with NBW_{Air}, a higher RA of *Firmicutes* can promote the hydrolysis and acidification of organic matter [28].

In comparison, another important member in the hydrolysis and acidification process is *Bacteroidetes*. *Bacteroidetes* are responsible for degrading cellulose and various proteins into fatty acids and ammonia [28]. On day 21, the RA of *Bacteroidetes* in the groups with added NBW is higher than DW, indicating that the addition of NBW promotes hydrolysis more effectively. The primary role of *Bacteroidetes* in AD is hydrolysis and acidification, so their RA is high in the early stage of AD and decreases with the end of hydrolysis [22]. *Firmicutes* and *Bacteroidetes* are closely associated with the hydrolysis and acidification of organic matter, with their RA showing a negative correlation [34]. During the AD process, the RA of *Firmicutes* increases, while the RA of *Bacteroidetes* decreases markedly.

Proteobacteria is a phylum often found in the AD process, and its main function is to decompose a large amount of organic matter and digest some VFA [35]. It plays a significant role in the late stage of digesting propionate in the digesters. As digestion progresses, the RA of *Proteobacteria* in the DW group decreases. Adding NBW_{Air}, NBW_{CO_2}, and NBW_{Air+CO_2} to the digesters can effectively increase the RA of *Proteobacteria*, enhance hydrolysis, and lead to an increase in methane yield. *Chloroflexi* is capable of degrading a wide range of complex macromolecules, including proteins and carbohydrates [36]. In the early stage of AD, the RA of *Chloroflexi* is between 2.5 %–5.4 %, and it rises to 6.9 %–12.7 % on day 21. This indicates that as AD progresses, the structure of the microbial community becomes more conducive to the enrichment of functional microorganisms, thereby accelerating the hydrolysis and acidification process of co-digestion with KW and SS. The dominant phylum of sludge in AD includes *Firmicutes*, *Bacteroidetes*, *Actinobacteria*, *Proteobacteria*, *Chloroflexi*, and *Spirochaetes*, with *Firmicutes* being the most prevalent in digesters such as restaurants and feces, which is consistent with the present study [37].

Bacteria were further analyzed at the genus level (**See supplementary material**). On day 3, the dominant genus in AD were *Streptococcus*, *Bacteroides*, and *Macellibacteroides*. On day 21, *Syntrophomonas*, and *Macellibacteroides* dominated. *Streptococcus* is primarily responsible for converting sugar to lactic acid [38]. The enrichment of *Streptococcus* in all digesters promoted the accumulation of acid. *Streptococcus* had a significantly negative correlation with methane yield ($p < 0.001$, R = -0.83), and the RA of *Streptococcus* in the digesters with NBW added was lower than that of DW on day 3, suggesting that NBW could enhance the metabolic activity of methanogens and thus increase methane yield. *Streptococcus* was less susceptible to the effects of AD, and the lower RA of *Streptococcus* in NBW digesters indicated that the addition of NBW was effective in mitigating the damage to the digesters by *Streptococcus* [39]. The decrease in the RA of *Streptococcus* on day 21 was probably due to the degradation of VFA in the digesters.

Syntrophomonas can be enriched upon contact with VFA and can establish a syntonic metabolism with methanogens [4]. Thus, *Syntrophomonas* was enriched during AD, with RA increasing to 20.2 %-27.2 %. There was a significantly positive correlation with CMY ($p < 0.001$, R = 0.75). *Bacteroides* perform an influential role in the hydrolysis of several polysaccharides and the production of VFA by AD [40]. It had a significantly positive correlation with VFA ($p < 0.001$, R = 0.78). On day 3, the RA of *Bacteroides* in NBW_{Air+CO_2} was 15.1 %, which was lower than the DW, which may explain the lesser VFA in the added NBW_{Air+CO_2} digesters. As digestion proceeded, the RA of *Bacteroides* gradually decreased and VFA was gradually consumed. *Macellibacteroides* can utilize monosaccharides to produce VFA, promote the conversion of proteins and carbohydrates to soluble sugars and VFA, and accelerate hydrolysis and acid production [41]. Its RA decreased from 9.2 %-11.9 % to 3.8 %-4.9 % during the AD process, when VFA was rapidly consumed.

PCA plot revealed differences between bacterial communities (Fig. 5**d**). At the hydrolysis stage, the addition of NBW differed from the DW, indicating that the addition of different NBW affected the activity of microorganisms. However, the difference was not significant in the methanogenic stage, which may be due to the slower change of the bacterial community in the late stage of AD [42].

3.3.3. The distributions of archaea sequences

Methanogens are archaea that perform an influential role in the carbon cycle in nature [43]. During the AD process, methanogens are more susceptible to external elements than bacteria, which may lead to changes in the function of archaea in response to changing environmental conditions [44]. The archaeal community at the phylum level consisted mainly of *Euryarchaeota* and *Nanoarchaeaota* (**See supplementary material**).

The RA of *Euryarchaeota* in the digesters (67.6 %-74.8 %) tended to increase with AD time in the same reactor (98.6 %-99.4 %). *Euryarchaeota* is a fastidious and strict anaerobe that utilizes CO_2 for energy production and carbon fixation via the Wood-Ljungdahl pathway (WLP) [45]. In this pathway, most bacteria (acetate-producing bacteria) produce acetate as an end product, while most archaea (CO_2-reducing methanogens) produce methane [46]. WLP application studies include (1) promotion of acetic acid production or chain elongation by injection of exogenous H_2 and CO_2, biogas recirculation, or addition of zero-valent iron to convert VFA to medium-chain fatty acids, and (2)

stimulation of CO_2 utilization by CO_2-reducing methanogens to drive methanogenic [47]. The presence of *Firmicutes*, *Proteobacteria* with *Euryarchaeota* in the digesters is justified by the fact that the product "hydrogen" released by them is used by the hydrogenotrophic methanogens in *Euryarchaeota* for methane yield. Therefore, *Firmicutes*, *Proteobacteria*, and *Euryarchaeota* should co-exist in the methane production process. *Nanoarchaeaota* is highly adaptable to harsh environments [2].

Further analyses were carried out at the genus level to gain insight into the structure of the archaeal community. In the early stage of AD, the digesters are dominated by three types of archaeal genus, *Methanobacterium*, *Methanocorpusculum,* and *Methanobrevibacter*. As AD proceeded, *Methanobacterium* became the only dominant genus (Fig. 6**a** and Fig. 6**b**).

Methanobacterium is a strict hydrogenotrophic methanogen, utilizing H_2 and CO_2 to produce methane [31]. Among all the digesters, *Methanobacterium* was enriched (53.5 %-56.6 %), and its RA increased to 92.1 %-96.1 % as digestion proceeded. The addition of NBW promoted the enrichment of *Methanobacterium* in the late stage of digestion, which was a significantly positive correlation with CMY ($p < 0.001$, R = 0.76) (Fig. 6**c**), suggesting that the addition of NBW could enhance the symbiosis between other bacteria and hydrogenotrophic methanogens, which in turn increased the methane yield. The RA of *Methanobacterium* was higher under microaerobic conditions, which was consistent with the present study [48].

Methanocorpusculum is commonly used in methanogenic bioreactor assays and is found in high RA in many leachates, where it can utilize H_2/CO_2 or formate as a substrate for methanogenesis [44]. In addition, *Methanocorpusculum* had a negative correlation with CMY ($p < 0.01$, R = -0.59), and its RA gradually decreased in the late stage of AD. *Methanobrevibacter* is also a hydrogenotrophic methanogenic bacterium with high acid tolerance [4]. The RA of Methanobrevibacter in the NBW-added digesters was higher on day 3, and had a positive correlation with VFA ($p < 0.001$, R = -0.79), suggesting that the introduction of NBW improved the hydrolysis of AD. As the VFA was consumed, the RA of *Methanobrevibacter* decreased to 0.3 %-0.5 %.

Methanomassiliiccoccus is a hydrogenotrophic and methylotrophic methanogen [2]. In the methanogenic stage, the RA of NBW_{Air+CO_2} was 1.0 %, and the other groups were less than 1.0 %, indicating that the enrichment of methanogens by hybrid NBW was more significant. In summary, a lower diversity of methanogenic flora was observed when the digesters treated KW and SS. This low diversity usually indicates a more sensitive and unstable ecosystem [49].

There are some differences in the PCA plot of the hydrolysis and methanogenic phases of the archaeal community (Fig. 6**d**), but the distances between the groups are similar, which is mainly due to the low diversity of the archaea.

4. Conclusion

The addition of NBW promoted hydrolysis and increased methane yield. The methane yield of NBW_{Air}, NBW_{CO_2}, and NBW_{Air+CO_2} increased by 4.7 %, 25.3 %, and 21.0 %. The addition of NBW_{Air+CO_2} improved the stability of digestion, decomposed organic matter most thoroughly, alleviated the inhibition caused by the accumulation of VFA, and facilitated the subsequent methanation process. The addition of NBW increased the RA of methanogens, which, together with hydrogenotrophic methanogens, improved the utilization of hydrogen, thereby increasing methane yield. In addition, NBW_{Air+CO_2} is inexpensive and has the potential for large-scale diffusion in industry.

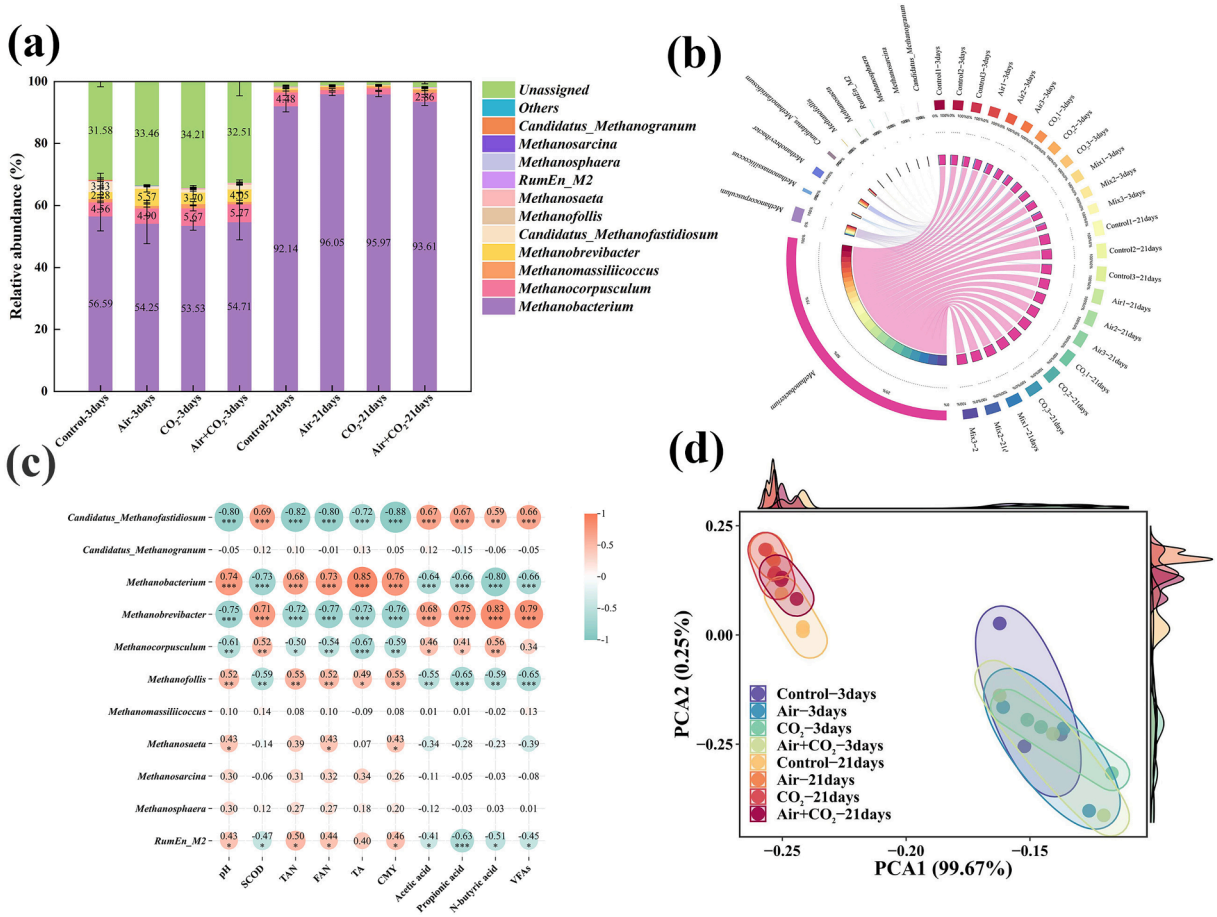


Fig. 6. Distribution of archaea sequences at the genus level, RA (a), Circos plot (b), Correlation analysis plots of environmental factors (c), and PCA plot (d).

CRediT authorship contribution statement

Qingfang Zhang: Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis. **Jiazi Niu:** Writing – original draft, Validation, Methodology, Investigation, Data curation. **Peilin Guo:** Methodology, Investigation, Data curation. **Jie Wang:** Software, Methodology, Investigation. **Cheng Peng:** Writing – review & editing, Software. **Yifan Zhang:** Validation. **Zhe Liu:** Software. **Yabian Wang:** Funding acquisition. **Liyang Xue:** Funding acquisition. **Tianfeng Wang:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2024.133456.

Data availability

Data will be made available on request.

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